

Coastal Engineering Tutorial 1

Problem One:

A fishing boat on the ocean surface finds itself moving up and down. They find that the vertical distance between the minimum and maximum elevation is 1 m and it takes 5 s for the boat to move from a trough to a crest. From looking at the movement of nearby buoys they estimate that the wavelength (from crest to crest) of the wave is about 200 m. From a depth sounder they know that the water depth is 600 m.

- A** Calculate the wave amplitude, wavenumber and angular frequency of the wave.
- B** Can the small wave approximation be used on this wave?
- C** Calculate the phase velocity based on the measurements made on the boat and the theoretical wavespeed based on the dispersion relation.
- D** Comment on the difference between the two phase velocities calculated in part C. Justify which phase velocity you think is more accurate and adjust the calculations of the basic wave properties in part A accordingly.

Problem Two:

A long wave signal is observed 200 km offshore from a populated beach. Where the wave was observed, the ocean depth is 2500 m. The measured wavelength is 275 km and the wave period is 1757 s. The amplitude of the wave is only 10 cm.

- A** Without making any calculations do you think the people on the beach should be concerned by this relatively small amplitude wave that's still 200 km offshore?
- B** Classify this wave as a shallow or a deep water wave based on the measured characteristics. Use this to calculate the group velocity (i.e. the velocity with which the wave energy will approach the shore) and how long it will take for the wave energy to reach the shore at its current velocity.
- C** Calculate the energy per wavelength of the wave. Compare this to a typical swell wave such as the one examined in the previous problem ($\eta_0 = 0.5$ m; $k = 0.0402$ rad/m; $\omega = 0.628$ rad/s)
- D** In practice the wave with 'shoal' as it approaches the shore causing its wavelength to decrease. To help compensate for this without actually calculating the shoaling directly, calculate the energy flux and the energy carried by a single wavelength for both the measured wave and our typical swell wave. You can still treat the wave as two-dimensional.

Problem Three: Describe the key assumptions that were used to derive our wave equations and which ones are the most restrictive.